

Minor- I

School of Mathematics and Statistics
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points

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Course code: ST 402

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

1. Let

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{bmatrix}$$

[5]

Find a row - reduced echelon matrix R which is row-equivalent to A and an invertible 3×3 matrix P such that $A = PR$.

2. Examine whether the following sets of vectors constitute a vector space. If so, find the dimension and basis of that vector space.

(a) The set of all real polynomials of degree 5 together with the zero polynomial.

[5]

(b) The set of all $n \times n$ real matrices which are symmetric.

[5]

3. Let V be an n -dimensional vector space over $\mathbb{R}$. Suppose B_1 and B_2 be two ordered bases of V . Then find **invertible** matrix P ($n \times n$) such that

$$[x]_{B_2} = P [x]_{B_1}$$

for every $x \in V$.

[5]

All the best !